

Is Human-AI Interaction CSCW?

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ABSTRACT

As AI systems progress in their capabilities, Human-AI Collaboration is likely to become an increasingly prevalent paradigm in many domains, including business, education, entertainment, and social activities. Historically, only computer-mediated collaborations involving at least two people have been considered part of the field of study known as CSCW (computer-supported cooperative work). This panel will engage CSCW 2024 attendees with the controversial and timely question of whether the purview of CSCW should be expanded to include research on "collaborations" involving a single person and one or more AI models or agents.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing systems and tools**; **Human computer interaction (HCI)**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

Artificial intelligence, Machine learning, Human-AI Collaboration, CSCW

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1 PANEL OVERVIEW

The scope of the discipline of "Computer-Supported Cooperative Work (CSCW)" and its associated ACM conference has evolved over time. Initially, as the nomenclature suggests, the field was focused on multi-user productivity applications, primarily related to knowledge work, such as enterprise email and calendaring tools [12, 27]. Over time, the scope of CSCW expanded to include multi-user computing systems more broadly, even when not in the context of

"work," including systems for video communications (whether for work or leisure), various types of computer-mediated communications, educational technologies, and even multi-user gaming [17–20, 22]. Even with this expanded set of topical domains, CSCW's focus was collaboration in pairs or small groups; however, the rise of internet technologies brought internet-scale collaborations into the scope of CSCW scholarship, including the study of technologies such as Wikipedia and crowdwork platforms [2, 13–15]. And, most recently, the advent of social media caused another broadening of scope (including a renaming of the ACM conference to append "and Social Computing" to the title, reflecting that social technologies now represented a far greater portion of CSCW scholarship than traditional, small-group collaboration technologies) [1, 6, 7, 10, 16]. In this panel, we consider the question of whether the field of CSCW should again expand its scope to include a growing and important technology trend – Human-AI Collaboration.

The rapid advancement and adoption of generative AI technologies including large language models [23] and multimodal models that generate images, video, audio, and/or code [25] is a defining trend of the 2020s, and one that is likely to continue. Indeed, many experts predict that Artificial General Intelligence (AGI) models that are as capable as humans across a wide spectrum of cognitive tasks might become available within the next decade [3, 4, 21]. Even if true "AGI" remains elusive, advancing capabilities of frontier models facilitate novel human-AI partnerships, such as with advanced AI assistants [9] and agent-based systems [24, 28], which often employ anthropomorphized personas. Advances in explainability (XAI) and interpretability of AI models [5, 26] are likely to further support complex human-AI interactions. While some academics argue that one can never truly "collaborate" with a computer [8, 11], the metaphor of human-AI collaboration is a powerful one that many companies, researchers, and end-users [9, 21, 29] find useful as a way to describe their subjective experiences of using frontier AI technologies and/or convey the affordances that such technologies offer.

In this panel, a group of academic and industry researchers at the forefront of building and studying Human-AI Interaction technologies will present a spectrum of viewpoints on the meaning of the term "Human-AI Collaboration," the types of scholarship needed to understand and support this phenomenon, and whether such work should have an academic home within the CSCW conference series.

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2 PANEL FORMAT

This panel will be structured as a moderated debate, in order to ensure coverage of a variety of viewpoints on this controversial topic, and to create an engaging session for the audience. Our five panel organizers will be divided into a moderator and two teams of two panelists each. The moderator will pose a series of resolutions (which will be provided to the teams beforehand for preparation). Teams will be randomly assigned in advance to support or oppose each resolution. For each resolution, the moderator will first poll the audience to understand the initial sentiment among attendees. Then, each team will have two minutes to provide an opening statement representing their position. They will then have two minutes to craft a one-minute rebuttal to the opposing teams' statement, which they will then present. We will then make the session interactive by having five minutes of questions from the audience relating to the resolution. Finally, the teams will have two minutes to each compose one-minute concluding remarks. After the concluding remarks, the moderator will take an audience vote to determine which side "won" the debate. We will prepare a total of five resolutions, but may do fewer as time permits during the session.

Below are several draft resolutions to seed the debate; if accepted, we will also elicit possible additional or alternate resolutions from CSCW attendees via social media before the event.

- The phrase "Human-AI Collaboration" is problematic because it misleads users about the agency of AI tools.
- We should model advanced AI systems as humans when designing human-AI interactions.
- Fully interpretable AI models are a necessary prerequisite to true human-AI collaboration.
- CSCW should expand its purview to include AIs as potential full partners in collaborative work because AIs are "close enough" to people in their capabilities (or soon will be).
- CSCW should expand its purview to include situations in which only AIs are interacting with each other in ways that seem collaborative or social and no humans are directly involved.

3 PANELISTS

Our panel consists of a moderator (Amy S. Bruckman) and four panelists: Meredith Ringel Morris, Jeffrey P. Bigham, Michael S. Bernstein, and Andrés Monroy-Hernández. These five individuals are all at the forefront of human-AI interaction research, have a long history of involvement with the discipline of CSCW, and represent a diversity of experiences and institutions, including experience in the startup world, experience working at "Big Tech" companies, and experience as university faculty.

Meredith Ringel Morris is Director for Human-AI Interaction Research at Google DeepMind. Prior to joining DeepMind, she served as Director of the People + AI Research (PAIR) team at Google Research. She was previously Research Area Manager for Interaction, Accessibility, and Mixed Reality at Microsoft Research. Dr. Morris is also an Affiliate Professor in the Paul G. Allen School of Computer Science & Engineering and The Information School at the University of Washington. She is an ACM Fellow and member of the SIGCHI Academy in recognition of her contributions to HCI. She served as General Co-Chair of CSCW 2016 and Program Co-Chair

of CSCW 2014, and was a member of the CSCW Steering Committee from 2012-2015. Her full bio and publications are available at <http://merrie.info>.

Michael S. Bernstein is an Associate Professor of Computer Science at Stanford University. He is currently serving as an Editor for CSCW 2024, and previously served as Program Co-Chair of UIST 2019, and Conference Co-Chair of HCOMP 2023 and Collective Intelligence 2023. His work introduced generative agents, jury learning, and other behavioral and systems work on human-AI interaction.

Jeffrey P. Bigham is Director of Human-Centered Machine Intelligence at Apple and an Associate Professor in the Human-Computer Interaction Institute in the School of Computer Science at Carnegie Mellon University. He has served as an AC for CSCW 2013 and CSCW 2015, and was a member of the Doctoral Consortium panel for CSCW 2016. He has also been the General Co-Chair of HCOMP 2014 and General Chair of ACM ASSETS 2019. At Carnegie Mellon University, he introduced the Human-AI Interaction course in 2018 and serves as the HCI representative for the SCS AI Major. His work in research and industry has generally advanced how people interact and benefit from machine learning, especially in the area of accessibility for people with disabilities. He is currently leading Apple's efforts in Responsible ML and AI Safety.

Andrés Monroy-Hernández is an Assistant Professor in Princeton University's Department of Computer Science, where he co-leads the Human-Computer Interaction Lab. Previously, he spent a decade in the tech industry. First at Microsoft Research, where he led the launch of Microsoft's first hybrid AI service, Calendar.help, a meeting scheduling bot part of Office. Then, at Snap, he was the founding research manager of the HCI research group, where he led the design of Snapchat's co-located social AR experiences. His current research involves designing and studying more equitable AI-mediated labor platforms, such as ride-share and food delivery. His service for CSCW includes being the technical program co-chair in 2018, editor and steering committee member from 2019 to 2021. His bio and publications are available at <https://andresmh.com>.

Amy S. Bruckman is Regents' Professor in the School of Interactive Computing in the College of Computing at the Georgia Institute of Technology. She does research on online collaboration, social movements, and content moderation, and is the author of the book *Should You Believe Wikipedia?* (Cambridge University Press, 2022). She is an ACM Fellow and member of the SIGCHI Academy. She was the general co-chair of CSCW 2013 and chair of the CSCW steering committee, and she is an Editor for CSCW 2024, where she has had to deal with the thorny question of whether research on human-AI interaction is a fit for CSCW.

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